05 MYMLP
Digital Transmission

TLC Virtual LAB#2

GOAL: build a numerical simulator to analyze the performance of a digital transmission system for PAM-M modulations (M=2,4, 8 and 16) with antipodal symbols in AWGN channels.

1. VALIDATION AGAINST THEORY
   Validate your simulator against the theoretical findings considering the MATCHED FILTER case for two different pulses: NRZ and Square Root Raised Cosine (SRRC).
   Properly select an adequate level of SpS and process enough bits to get accurate validation.
   Required plots: BER vs. E_b/N_0, power spectral density and eyediagram (noiseless and at Eb/N0 equivalent to BER=10^{-3}).

2. PERFORMANCE OF A RECEIVER BASED ON A SINGLE-POLE FILTER for PAM-M: BANDWIDTH OPTIMIZATION
   Consider both NRZ and SRRC shaping at transmitter side.
   Optimize the single-pole filter bandwidth: determine the bandwidth of the filter that minimize the E_b/N0 required to deliver BER=10^{-3}.
   Determine the E_b/N_0 penalty of the best single-pole filter compared to the matched filter.
   Required plots: E_b/N_0 @ BER=10^{-3} vs. filter -3 dB bandwidth and eyediagram for optimal filters (noiseless and at Eb/N0 equivalent to BER=10^{-3}).

PRELIMINARY TASKS: DIGITAL FILTER SYNTHESIS

I. SINGLE POLE FILTER SYNTHESIS THROUGH IMPULSE INVARIANCE METHOD
II. SRRC FILTER SYNTHESIS THROUGH FREQUENCY SAMPLING METHOD

In both case you must provide evidence, in the spectral domain, of the good quality of the digital synthesis in the frequency region occupied by the signal.